

Justin Rosales

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EDUCATION

University of Michigan

BSE Computer Science, Minor Mathematics

Ann Arbor, MI

Expected May 2028

- **GPA:** 3.8/4.0

- **Coursework:** Data Structures and Algorithms, Computer Organization, OOP and Intro Data Structures

EXPERIENCE

Student Software Developer

May 2026 – Present

University of Michigan School of Nursing

Ann Arbor, MI

- Engineered a dynamic 11-step clinical assessment and diagnostic HADS screening tool in React Native, using state-driven branching logic for instant patient anxiety scoring and clinical severity feedback
- Designed resilient client-side state management using Redux Toolkit slices and async dispatchers, achieving zero runtime crashes in offline hospital environments with automated mock fallbacks
- Integrated assessment outcomes into a dynamic journey tracking and resource recommendation engine via RESTful MongoDB endpoints, automating milestone logging and filtering clinical articles tailored to user state

Research Assistant

May 2026 – Present

University of Michigan – Civil & Environmental Engineering Department

Ann Arbor, MI

- Architected the migration of monolithic Jupyter Notebooks into modular, production-grade Python pipelines for aerosol mass spectrometry (AMS) analysis, eliminating codebase redundancy across CLI and GUI interfaces
- Engineered a responsive desktop GUI application in CustomTkinter with embedded interactive Matplotlib canvases for real-time functional group deconvolution, reducing analytical visualization workflow time by 80%
- Developed automated batch-processing data pipelines with error handling to sequentially analyze multi-file atmospheric chemistry datasets, scaling throughput to generate 100+ publication-quality radial plots in seconds

PROJECTS

Arithmetic Battle | [Live Demo](#) | [GitHub](#) | *React, TypeScript, Tailwind CSS, Firebase*

2026

- Developed a real-time multiplayer mental math duel game with head-to-head matchmaking, enabling zero-friction player entry via Firebase Anonymous Auth and instant room creation with 6-character codes
- Synchronized live opponent scoring and game states with low latency via Firebase Firestore real-time document listeners, while implementing a multiplayer rematch voting system with immediate transactional state resets
- Ensured fair gameplay with identical mathematical problem sequences and arithmetic difficulty across both client instances by implementing a deterministic seeded pseudo-random number generation algorithm (Mulberry32)

Scratchpad | [Live Demo](#) | [GitHub](#) | *React, Node.js, Express, MongoDB, Redis, Tailwind CSS*

2026

- Architected a MERN stack notes application with client-side SPA capabilities using React, Express, and Mongoose ODM mapping to MongoDB, achieving 100% data integrity and sub-second updates across all CRUD transactions
- Secured public REST API endpoints against DoS threats and bot abuse by integrating Upstash Redis rate-limiting middleware in Express, dynamically throttling excess per-IP traffic with automated 429 error responses
- Designed a responsive UI with Tailwind CSS and DaisyUI components, implementing custom deletion confirmation modals and Express wildcard fallback routing to eliminate client-side SPA navigation errors in production

Splitr | *MHacks 2025 Winner – Best Use of Auth0* | [Devpost](#) | *Flask, Auth0, OpenCV, PyTorch, SQLite*

2025

- Developed a secure web application featuring real-time facial recognition via OpenCV, PyTorch, and FaceNet to identify users through live webcam video feeds, achieving over 90% biometric identity verification accuracy
- Integrated secure authentication and session management using Flask protected route decorators, leveraging Auth0 to implement OAuth 2.0 security with social sign-in, passwordless login, and multi-factor support
- Co-designed, engineered, and successfully deployed the functional prototype within a 24-hour hackathon time limit in a 4-person collaborative engineering team, winning 1st Place for “Best Use of Auth0” at MHacks 2025

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C/C++, HTML, CSS

Frameworks/Libraries: React, Tailwind CSS, NumPy, pandas

Backend/Databases: Node.js, Express, MongoDB, SQLite

Tools/Other: Git, Vercel, Firebase, Neovim, Antigravity, VS Code